Assignment 3 - Part 1 - Report

Natalie Poche

Part 1: Bandit_3

Level 12:

```
bandit11@bandit: ~

bandit11@bandit:~$ ls -al
total 24
drwxr-xr-x   2 root     root     4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root     4096 Apr  3 15:20 ..
-rw-r--r--   1 root     root      220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root     3851 Apr  3 15:10 .bashrc
-rw-r-----   1 bandit12 bandit11   49 Apr  3 15:17 data.txt
-rw-r--r--   1 root     root      807 Mar 31  2024 .profile
bandit11@bandit:~$ cat data.txt
Gur cnffjbeq vf 7k16JArUVv5LxVuJfsSVdbbtaHGlw9D4
bandit11@bandit:~$ tr 'A-Za-z' 'N-ZA-Mn-za-m' < data.txt
The password is 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4
bandit11@bandit:~$
```

7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4

```
bandit12@bandit: ~

PS C:\Users\Owner> ssh bandit12@bandit.labs.overthewire.org -p 2220
                      _                     _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit12@bandit.labs.overthewire.org's password:

      ,----..            ,----,          .---.
     /   /   \         ,/   .`|         /. ./|
    /   .     :      ,`    .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :     | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `____'  |  | \   ;  `      |
  '   ;  \; /  |     '   :  ;   .   \    .\  ;
   \   \  ',  /      |   |  '    \   \   ' \ |
    ;   :    /       '   :  |     :   '  |--"
     \   \ .'        ;   |.'       \   \ ;
  www. `---` ver     '---' he       '---" ire.org


Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

Used the ls command to list the directory contents, and the flag -a tells it to include hidden files, such as those that start with a dot, and the -l flag formats the output as a detailed list showing permissions, ownership, and sizes, which was used to discover data.txt, which was then concatenated and printed to standard output which revealed that the file contains encrypted text. The tr command is used to translate or transliterate characters, the specific commands read from data.txt, which is shown from the "<" to indicate input redirection, and maps every letter to the standard alphabet, which is shown through the "A-Za-z" to a character shifted by 13

places like it mentioned in the instructions which I shown through "N-ZA-Mn-za-m", which decides the ciphertext to reveal the password.

Level 13:

```
bandit12@bandit: /tmp/tmp.e
bandit12@bandit:~$ ls -al
total 24
drwxr-xr-x   2 root     root     4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root     4096 Apr  3 15:20 ..
-rw-r--r--   1 root     root      220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root     3851 Apr  3 15:10 .bashrc
-rw-r-----   1 bandit13 bandit12 2575 Apr  3 15:17 data.txt
-rw-r--r--   1 root     root      807 Mar 31  2024 .profile
bandit12@bandit:~$ xxd -r < data.txt | file -
/dev/stdin: gzip compressed data, was "data2.bin", last modified: Fri Apr  3 15:17:20 2026, max compression, from Unix
bandit12@bandit:~$ xxd -r < data.txt | zcat | file -
/dev/stdin: bzip2 compressed data, block size = 900k
bandit12@bandit:~$ xxd -r < data.txt | zcat | bzip2 --decompress --force --stdout | file -
/dev/stdin: gzip compressed data, was "data4.bin", last modified: Fri Apr  3 15:17:20 2026, max compression, from Unix
bandit12@bandit:~$ xxd -r < data.txt | zcat | bzip2 --decompress --force --stdout | zcat | file -
/dev/stdin: POSIX tar archive (GNU)
bandit12@bandit:~$ cd $(mktemp -d)
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ xxd -r < $HOME/data.txt | zcat | bzip2 --decompress --force --stdout | zcat | tar -xf -
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ file *
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ tar xvf data5.bin
data6.bin
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ file *
data5.bin: POSIX tar archive (GNU)
data6.bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ tar xvf data6.bin
data8.bin
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ file *
data5.bin: POSIX tar archive (GNU)
data6.bin: bzip2 compressed data, block size = 900k
data8.bin: gzip compressed data, was "data9.bin", last modified: Fri Apr  3 15:17:20 2026, max compression, from Unix, original size modulo 2^32 49
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$ zcat data8.bin
The password is FO5dwFsc0cbaIiH0h8J2eUks2vdTDwAn
bandit12@bandit:/tmp/tmp.e9nwEvkI4x$
```

FO5dwFsc0cbaIiH0h8J2eUks2vdTDwAn

```
bandit13@bandit: ~

PS C:\Users\Owner> ssh bandit13@bandit.labs.overthewire.org -p 2220
                      _                     _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit13@bandit.labs.overthewire.org's password:

      ,----..            ,----,          .---.
     /   /   \         ,/   .`|         /. ./|
    /   .     :      ,`    .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :    | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  | \   ;  `      |
  '   ;  \; /  |      '   :  ;   .   \    .\  ;
   \   \   ',  /       |   |  '    \   \   ' \ |
    ;   :    /        '   :  |     :   '  |--"
     \   \ .'         ;   |.'       \   \ ;
  www. `---` ver      '---' he       '---" ire.org


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--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```
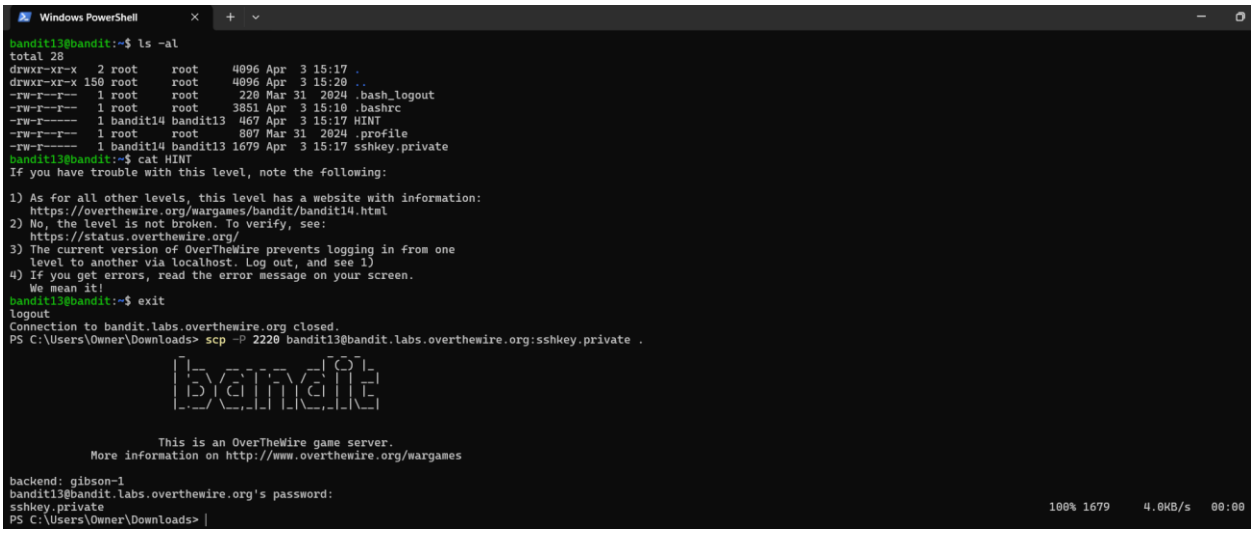

Because the instructions said the data.txt was a hex dump file, once it was confirmed to be in the directory using ls -al, xxd was then used since it is used to create or reverse hex dumps, with the -r flag to indicate that the data.txt file is to be reversed back into binary data. The pipe "|" sends the binary data to the file – command to analyze the standard input rather than a file on a disk to reveal the first layer of compression is gzip. Afterwards, xxd -r data.txt | zcat | file – was added, where zcat is used to decompress the gzip layer from data.txt and that decompression is pushed into the file – command to reveal that the second layer of compression is bzip2, which was solved by using bzip2 which is used as a file compression

program that uses the Burrows-Wheeler block sorting text compression algorithm, which the flag –decompress flag tells the program to decompress the incoming data, -- force flag tells the program to overwrite output files if they exist and bypass some of the standard safety checks, --stdout flag forces the decompressed data to be written to standard output so it can be passed into the next pipe rather than trying to save it as a file on the disk, which then reveals that the next layer is another gzip compression so the zcat is added next in the pipleine once again after the bzip2 commands and then passed into the file – pipe which shows that the next layer is a tar file that needs to be extracted, which mktemp -d is then used to create a secure temprary directory in /tmp and $() takes the outptu directory path and passes it to cd which allows the user to move into a writable workspace in order to safely extract files where data.txt had $HOME/ added to it since we needed to navigate to the directory the data.txt file was in from our current directory, then tar -xf – is then added to the pipleline to be used to extract the file where -x is the flag to extract, flag -f - is the flag to read the archive from standard input, file * is then used to show the contents of whatever file was extracted which revealed that there was another file to be extracted which tar xvf databin5.bin was used to extract the tarball with the -v flag to make the extraction verbose, listing the files as they are pulled out, file * was then used to view what file was extracted which revealed that data6.bin was another file that needed to have a file extracted from it so tar xvf data6.bin was then used to extract the next file and file * was used to view the extracted file which revealed that it was compressed with a gzip so zcat data8.bin was used to decompress the gzipped file which then revealed the password in the terminal.

Level 14:



-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAxkkOE83W2cOT7IWhFc9aPaaQmQDdgzuXCv+ppZ
Ha++buSkN+gg0tcr7Fw8NLGa5+Uzec2rEg0WmeevB13AIoYp0MZyETq46
t+jk9puNwZwIt9XgBZufGtZEwWbFWw/vVLNwOXBe4UWStGRWzgPpEeS
v5Tb1VjLZIBdGphTIK22Amz6ZbThMsiMnyJafEwJ/T8PQO3myS91vUHEu
oOMAzoUID4kN0MEZ3+XahyK0HJVq68KsVObefXG1vvA3GAJ29kxJaqvR

fgYnqZryWN7w3CHjNU4c/2Jkp+n8L0SnxaNA+WYA7jiPyTF0is8uzMlYQ4I
1Lzh/8/MpvhCQF8r22dwIDAQABAoIBAQC6dWBjhyEOzjeAJ3j/RWmap9M
5zfJ/wb2bfidNpwbB8rsJ4sZIDZQ7XuIh4LfygoAQSS+bBw3RXvzEpvJt3Sm
U8hIDuLsCjL1VnBY5pY7Bju8g8aR/3FyjyNAqx/TLfzlLYfOu7i9Jet67xAh0tO
NG/u8FB5I3LAI2Vp6OviwvdWeC4nOxCthldpuPKNLA8rmMMVRTKQ+7T2
VSnXmwYckKUcUgzoVSpiNZaS0zUDypdpy2+tRH3MQa5kqN1YKjvF8RC
47woOYCktsDo3FFpGNFec9Taa3Msy+DfQQhHKZFKIL3bJDONtmrVvtYK
40/yeU4aZ/HA2DQzwheol1AfiEhAoGBAOnVjosBkm7sblK+n4IEwPxs8sO
mhPnTDUy5WGrpSCrXOmsVIBUflaL3ZGLx3xCIwtCnEucB9DvN2HZkupc/
h6hTKUYLqXuyLD8njTrbRhLgbC9QrKrSM1F2fSTxVqPtZDlDMwjNR04xH
A/fKh8bXXyTMqOHNJTHHNhbh3McdURjAoGBANkU1hqfnw7+aXncJ9bjy
sr1ZWbqOE5Nd8AFgfwaKuGTTVX2NsUQnCMWdOp+wFak40JHPKWkJN
dBG+ex0H9JNQsTK3X5PBMAS8AfX0GrKeuwKWA6erytVTqjOfLYcdp5+z
9s8DtVCxDuVsM+i4X8UqIGOlvGbtKEVokHPFXP1q/dAoGAcHg5YX7WEe
hCgCYTzpO+xysX8ScM2qS6xuZ3MqUWAxUWkh7NGZvhe0sGy9iOdANz
wKw7mUUFViaCMR/t54W1GC83sOs3D7n5Mj8x3NdO8xFit7dT9a245Tvao
YQ7KgmqpSg/ScKCw4c3eiLava+J3btnJeSIU+8ZXq9XjPRpKwUCgYA7z6
LiOQKxNeXH3qHXcnHok855maUj5fJNpPbYiDkyZ8ySF8GlcFsky8Yw6fW
CqfG3zDrohJ5l9JmEsBh7SadkwsZhvecQcS9t4vby9/8X4jS0P8ibfcKS4nBP
+dT81kkkg5Z5MohXBORA7VWx+ACohcDEkprsQ+w32xeDqT1EvQKBgQ
DKm8ws2ByvSUVs9GjTilCajFqLJ0eVYzRPaY6f++Gv/UVfAPV4c+S0kAW
pXbv5tbkkzbS0eaLPTKgLzavXtQoTtKwrjpolHKIHUz6Wu+n4abfAIRFubOd
N/+aLoRQ0yBDRbdXMsZN/jvY44eM+xRLdRVyMmdPtP8belRi2E2aEzA==
-----END RSA PRIVATE KEY-----

```
bandit14@bandit: ~

PS C:\Users\Owner\Downloads> ssh -i .\sshkey.private bandit14@bandit.labs.overthewire.org -p 2220
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_|_| |_|\__,_|_|\__|


                       This is an OverTheWire game server.
             More information on http://www.overthewire.org/wargames

backend: gibson-1

       ,----..            ,----,          .---.
     /   /   \         ,/   .`|         /. ./|
    /   .     :      ,`    .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :    | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  | \   ;  `      |
  '   ;  \; /  |     '   :  ;   .   \    .\  ;
   \   \  ',  /      |   |  '    \   \   ' \ |
    ;   :    /       '   :  |     :   '  |--"
     \   \ .'         ;   |.'       \   \ ;
  www. `---` ver      '---' he       '---" ire.org


Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
    * PASSWORDS for each level are stored in /etc/somegame_pass/.
```

The command ls -al was used to view all the files and file information in the directory, used cat HINT to view the contents of the HINT file, based on the file it shows that there was an ssh privte key that could only be accessed by the user bandit14 and that it was the password to get to level 14, so scp was used as it was a command to secure the copy protocol, -P 2220 was used to specify the port since it was not a specified port, bandit13@bandit.labs.overthewire.org:sshkey.private was used to specify the file location that was being copied over for and the trailing. dictates that the file should be saved in the current local directory. Once the file was obtained, ssh was then used to connect to the next level to remotely since the command establishes a secure encrypted connection to remote servers, the -i flag is used to specify that a key should be used and .\sshkey.private is used to specify that file to be used as the RSA private key for authentiction instead of asking for the password, bandit14@bandit.labs.overthewire.org was used to specify where we are trying to remotely get into and -p 2220 is the nonconventional port that the it is from.

Level 15:

```
bandit14@bandit: ~
bandit14@bandit:~$ ls -al
total 24
drwxr-xr-x   3 root root 4096 Apr  3 15:17 .
drwxr-xr-x 150 root root 4096 Apr  3 15:20 ..
-rw-r--r--   1 root root  220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root root 3851 Apr  3 15:10 .bashrc
-rw-r--r--   1 root root  807 Mar 31  2024 .profile
drwxr-xr-x   2 root root 4096 Apr  3 15:17 .ssh
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14
MU4VWeTyJk8ROof1qqmcBPaLh7lDCPvS
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14 | nc localhost 30000
Correct!
8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo

bandit14@bandit:~$
```

8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo

```
bandit15@bandit: ~

PS C:\Users\Owner\Downloads> ssh bandit15@bandit.labs.overthewire.org -p 2220
                      _                     _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit15@bandit.labs.overthewire.org's password:

       ,----..            ,----,          .---.
     ,/   /   \         ,/   .`|         /. ./|
    /   .     :      ,`    .'  :     .--'.  ' ;
   .   /   ;.  \   ;     ;     /    /__./ \ : |
  .   ;   /  ` ; .'---,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :    | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  | \   ;  `      |
  '   ;  \; /  |     '   :  ;   .   \    .\  ;
   \   \  ',  /      |   |  '    \   \   ' \ |
    ;   :    /       '   :  |     :   '  |--"
     \   \ .'        ;   |.'       \   \ ;
  www. `---` ver     '---' he       '---" ire.org


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--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

Cat /etc/bandit_pass/bandit14 reads the current level's password and uses "|" to pipe it into the netcat (nc) command which is used to open a TCP connection to the service running on localhost port 30000 as was specified in the over the wire instructions, this then transmits the password and prints the server's response which is the next password into the terminal.

Level 16:

```
bandit15@bandit: ~

bandit15@bandit:~$ ls -al
total 24
drwxr-xr-x   2 root     root     4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root     4096 Apr  3 15:20 ..
-rw-r-----   1 bandit15 bandit15   33 Apr  3 15:17 .bandit14.password
-rw-r--r--   1 root     root      220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root     3851 Apr  3 15:10 .bashrc
-rw-r--r--   1 root     root      807 Mar 31  2024 .profile
bandit15@bandit:~$ echo "8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo" | openssl s_client -quiet -connect localhost:30001
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
Correct!
kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx

bandit15@bandit:~$ |
```

kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx

```
bandit16@bandit: ~

PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh bandit16@bandit.labs.overthewire.org -p 2220

                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit16@bandit.labs.overthewire.org's password:

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  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

Put the previous password to create a simple output message to be piped to retrieve the next credential using echo with the previous password, which is then piped into the openssl s_client command to negotiate the SSL handshake -quiet flag is then used to supress the extra certificate information, where -connect localhost:30001 tells where to connect to which is taken from the instructions.

Level 17:

```
bandit16@bandit: ~        Windows PowerShell

bandit16@bandit:~$ ls -al
total 24
drwxr-xr-x   2 root     root     4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root     4096 Apr  3 15:20 ..
-rw-r-----   1 bandit16 bandit16   33 Apr  3 15:17 .bandit15.password
-rw-r--r--   1 root     root      220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root     3851 Apr  3 15:10 .bashrc
-rw-r--r--   1 root     root      807 Mar 31  2024 .profile
bandit16@bandit:~$ nmap -sV localhost -p 31000-32000
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-21 10:53 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE     VERSION
31046/tcp open  echo
31518/tcp open  ssl/echo
31691/tcp open  echo
31790/tcp open  ssl/unknown
31960/tcp open  echo
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port31790-TCP:V=7.94SVN%T=SSL%I=7%D=4/21%Time=69E7574A%P=x86_64-pc-linu
SF:x-gnu%r(GenericLines,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x2
SF:0current\x20password\.\n")%r(GetRequest,32,"Wrong!\x20Please\x20enter\x
SF:20the\x20correct\x20current\x20password\.\n")%r(HTTPOptions,32,"Wrong!\
SF:x20Please\x20enter\x20the\x20correct\x20current\x20password\.\n")%r(RTS
SF:PRequest,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x20current\x20
SF:password\.\n")%r(Help,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x
SF:20current\x20password\.\n")%r(FourOhFourRequest,32,"Wrong!\x20Please\x2
SF:0enter\x20the\x20correct\x20current\x20password\.\n")%r(LPDString,32,"W
SF:rong!\x20Please\x20enter\x20the\x20correct\x20current\x20password\.\n")
SF:%r(SIPOptions,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x20curren
SF:t\x20password\.\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 167.73 seconds
bandit16@bandit:~$ echo "kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx" | ncat --ssl localhost 31790
Correct!
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAvmOkuifmMg6HL2YPIOjon6iWfbp7c3jx34YkYWqUH57SUdyJ
imZzeyGC0gtZPGujUSxiJSWI/oTqexh+cAMTSMlOJf7+BrJObArnxd9Y7YT2bRPQ
Ja6Lzb558YW3FZl87ORiO+rW4LCDCNd2lUvLE/GL2GWyuKN0K5iCd5TbtJzEkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WWAW1BBR6yGrMq7Q/kALHYW3OekePQAzL0VUYbW
JGTi65CxbCnzc/w4+mqQyvmzpWtMAzJTzAzQxNbkR2MBGySxDLrjg0LWN6sK7wNX
x0YVztz/zbIkPjfkU1jHS+9EbVNj+D1XFOJuaQIDAQABAoIBABagpxpM1aoLWfvD
kHcj10nqcoBc4oE11aFYQwik7xfW+24pRNuDE6SFthOar69jp5RlLwD1NhPx3iBl
J9nOM8OJ0VToum43UOS8YxF8WwhXriYGnc1sskbwpXOUDc9uX4+UESzH22P29ovd
d8WErY0gPxun8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQnPw9nC
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52yOQ9qOkwFTEQpjtF4uNtJom+asvlpmS8A
vLY9r60wYSvmZhNqBUrj7lyCtXMIu1kkd4w7F77k+DjHoAXyxcUp1DGL51sOmama
+TOWWgECgYEA8JtPxP0GRJ+IQkX262jM3dEIkza8ky5moIwUqYdsx0NxHgRRhORT
8c8hAuRBb2G82so8vUHk/fur85OEfc9TncnCY2crpoqsghifKLxrLgtT+qDpfZnx
SatLdt8GfQ85yA7hnWWJ2MxF3NaeSDm75Lsm+tBbAiyc9P2jGRNtMSkCgYEAypHd
HCctNi/FwjulhttFx/rHYKhLidZDFYeiE/v45bN4yFm8x7R/b0iE7KaszX+Exdvt
SghaTdcG0Knyw1bpJVyusavPzpaJMjdJ6tcFhVAbAjm7enCIvGCSx+X3l5SiWg0A
R57hJglezIiVjv3aGwHwvlZvtszK6zV6oXFAu0ECgYAbjo46T4hyP5tJi93V5HDi
Ttiek7xRVxUl+iU7rWkGAXFpMLFteQEsRr7PJ/lemmEY5eTDAFMLy9FL2m9oQWCg
R8VdwSk8r9FGLS+9aKcV5PI/WEKlwgXinB3OhYimtiG2Cg5JCqIZFHxD6MjEGOiu
L8ktHMPvodBwNsSBULpG0QKBgBAplTfC1HOnWiMGOU3KPwYWt0O6CdTkmJOmL8Ni
blh9elyZ9FsGxsgtRBXRsqXuz7wtsQAgLHxbdLq/ZJQ7YfzOKU4ZxEnabvXnvWkU
YOdjHdSOoKvDQNWu6ucyLRAWFuISeXw9a/9p7ftpxm0TSgyvmfLF2MIAEwyzRqaM
77pBAoGAMmjmIJdjp+Ez8duyn3ieo36yrttF5NSsJLAbxFpdlc1gvtGCWW+9Cq0b
dxviW8+TFVEBl1O4f7HVm6EpTscdDxU+bCXWkfjuRb7Dy9GOtt9JPsX8MBTakzh3
vBgsyi/sN3RqRBcGU40fOoZyfAMT8s1m/uYv52O6IgeuZ/ujbjY=
-----END RSA PRIVATE KEY-----
```

```
PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh -i sshkey17.private bandit17@bandit.labs.overthewire.org -p 222
0
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1

      ,----..            ,----,          .---.
     /   /   \        ,/   .`|         /. ./|
    /   .     :      ,`   .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ;  .   ___,/    ' .--'.  '   \' .
  ;   |  ; \ ; | |  .        / /___/ \ |    ' '
  |   :  | ; | ' ;  |.';  ; ;   \  \;      :    ;
  .   |  ' ' ' : `----'  |  | \   ;  `      |    |
  '   ;  \; /  |      '  :  ;  .   \    .\  ;    ;
   \  \  ',  /       |  |  '    \  \   ' \ |    |
    ;  :    ' /       '  :  |     :  '   |--"
     \  \   \ .'        ;  |.'       \  \  ;
  www. `---` ver     '---' he        '---" ire.org


Welcome to OverTheWire!

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discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

Based on the instructions nmap is used to scan the network as it is used to identify open ports the flag -sV probes the open ports to determine the service and version of info running on them localhost specifies the IP to look in, the flag -p 31000-32000 is the range of ports for the nmap scan to look through which is narrowed down by the given bandit instructions, this was used to

find the specific port range that was running the necessary SSL service, once it was found, echo with the password was used to pipe into ncat to establish a secure connection with the flag –ssl to establish a secure connection while also specifying localhost and the port number to get the correct port from the scan to get the RSA private key, which was copy and pasted into a sshkey17.private file to be used as the private key to use instead of a password for the next level through ssh -i ./sshkey17.private bandit17@bandit.labs.overthewire.org -p 2220.

Level 18:

```
bandit16@bandit: ~        bandit17@bandit: ~
bandit17@bandit:~$ ls -al
total 36
drwxr-xr-x   3 root     root     4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root     4096 Apr  3 15:20 ..
-rw-r-----   1 bandit17 bandit17   33 Apr  3 15:17 .bandit16.password
-rw-r--r--   1 root     root      220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root     3851 Apr  3 15:10 .bashrc
-rw-r-----   1 bandit18 bandit17 3300 Apr  3 15:17 passwords.new
-rw-r-----   1 bandit18 bandit17 3300 Apr  3 15:17 passwords.old
-rw-r--r--   1 root     root      807 Mar 31  2024 .profile
drwxr-xr-x   2 root     root     4096 Apr  3 15:17 .ssh
bandit17@bandit:~$ diff -u passwords.old passwords.new
--- passwords.old       2026-04-03 15:17:25.580141500 +0000
+++ passwords.new       2026-04-03 15:17:25.584253266 +0000
@@ -39,7 +39,7 @@
 RbNK7oUO0l50r8xbQFBMngXjP4PosdxI
 tA9dxz3SyAam2tiruMl4mB5CocAYTBP9
 I9IZK0CIWzeGdik4YD9bATBufq1fr9ON
-KxOU4IzbXM8j8HeAWPAXTd1eC77mp1qV
+x2gLTTjFwMOhQ8oWNbMN362QKxfRqGlO
 SlNJjK3d31Mr3jtNETeMLcrZ7Yx2izqR
 c6xvvFLhpp6EsauHO5E5I2k5Jmslv49K
 4kHcF6SFl6wZQIQofBBlaFjvjL89X4UH
bandit17@bandit:~$
```

x2gLTTjFwMOhQ8oWNbMN362QKxfRqGlO

```
bandit16@bandit: ~        Windows PowerShell
PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh bandit18@bandit.labs.overthewire.org -p 2220
                      _                                  _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit18@bandit.labs.overthewire.org's password:

      ,----..            ,----,          .---.
     /   /   \         ,/   .`|         /. ./|
    /   .     :      ,`   .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :    | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  | \   ;  `      |
  '   ;  \; /  |      '   :  ;   .   \    .\  ;
   \  \  ',  /       |   |  '    \   \   ' \ |
    ;  :    /        '   :  |     :   '  |--"
     \  \  .'         ;   |.'       \   \ ;
  www. `---` ver      '---' he        '---" ire.org


Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

The command ls -al was used to view all the files and their information in the directory to verify that the correct files from the instructions, passwords.old and passwords.new were present, once that was confirmed, the diff command was used to compare two files line by line the -u flag was used so the differences were outputted in a unified format and providing context lines around the change, and the command follows the two files to compare, because the instructions stated that the password is only the modified line, the password is found with the string that had the "+" sign by it.

Level 19:

```
Windows PowerShell        Windows PowerShell
PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh bandit18@bandit.labs.overthewire.org -p 2220 -t /bin/sh
                      _                                  _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit18@bandit.labs.overthewire.org's password:
$ pwd
/home/bandit18
$ ls -al
total 24
drwxr-xr-x   2 root     root     4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root     4096 Apr  3 15:20 ..
-rw-r--r--   1 root     root      220 Mar 31  2024 .bash_logout
-rw-r-----   1 bandit19 bandit18 3874 Apr  3 15:17 .bashrc
-rw-r--r--   1 root     root      807 Mar 31  2024 .profile
-rw-r-----   1 bandit19 bandit18   33 Apr  3 15:17 readme
$ cat readme
cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8
$
```

cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8

```
Windows PowerShell    bandit19@bandit: ~
PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh bandit19@bandit.labs.overthewire.org -p 2220
                      _                     _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit19@bandit.labs.overthewire.org's password:

       ,----..            ____          .---.
     /   /   \         ,/   .`|        /. ./|
    /   .     :      ,`   .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .__'.  '   \' .
  ;   |  ; \ ; | |    :    | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;   \  \;      :
  .   |  ' ' ' : `----'  |  | '\   ;  `      |
  '   ;  \; /  |     '   :  ;  .   \    .\  ;
   \   \   ',  /      |   |  '    \   \   ' \ |
    ;   :    /       '   :  |     :   '  |--"
     \   \ .'        ;   |.'       \   \ ;
  www. `---` ver     '---' he       '---" ire.org


Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

Based on the instructions the .bashrc file was modified for the specific user that contains a command that forces an immediate logout, so -t /bin/sh was added onto the usual ssh login information where the -t flag forces the ssh client to allocate a pseudo-terminal and launch the shell /bin/sh directly which was determined by the instructions to bypass the malicious script and grant the terminal access. The pwd command was used to confirm that we were logged in as the correct user, as it is used to print the working directory, then ls -al was used to locate the readme file that was specified in the instructions to have the password, once the readme file was located, the cat command to print the content in the terminal to reveal the password.

Level 20:

```
Windows PowerShell    bandit19@bandit: ~
bandit19@bandit:~$ ls -al
total 36
drwxr-xr-x   2 root     root      4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root      4096 Apr  3 15:20 ..
-rwsr-x---   1 bandit20 bandit19 14888 Apr  3 15:17 bandit20-do
-rw-r--r--   1 root     root       220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root      3851 Apr  3 15:10 .bashrc
-rw-r--r--   1 root     root       807 Mar 31  2024 .profile
bandit19@bandit:~$ file bandit20-do
bandit20-do: setuid ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux.so.2, BuildID[sha1]=d9b51170e04a
f1b03902f80228afa7a973330f86, for GNU/Linux 3.2.0, not stripped
bandit19@bandit:~$ ./bandit20-do
Run a command as another user.
  Example: ./bandit20-do whoami
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO
bandit19@bandit:~$
```

0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO

```
Windows PowerShell    bandit20@bandit: ~
PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh bandit20@bandit.labs.overthewire.org -p 2220
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                       This is an OverTheWire game server.
             More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit20@bandit.labs.overthewire.org's password:

      ,----..            ,----,          .---.
     /   /   \         ,/   .`|         /. ./|
    /   .     :      ,`   .'  :     .--'.  ' ;
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   ;   /  ` ; .'___,/    ,' .--'.  '   \' .
  ;   |  ; \ ; | |    :     | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;    \  \;      :
  .   |  ' ' ' : `----'  |  | \   ;  `      |
  '   ;  \; /  |      '   :  ;   .   \    .\  ;
   \   \  ',  /       |   |  '    \   \   ' \ |
    ;   :    /        '   :  |     :   '  |--"
     \   \ .'         ;   |.'       \   \ ;
  www. `---` ver      '---' he       '---"  ire.org


Welcome to OverTheWire!

If you find any problems, please report them to the #wargames channel on
discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

The command ls -al is used to view all the files and their information in the directory, file bandit20-do is used to reveal that bandit20-do is a setuid executable. This means that when the program is run, it executes with the privileges of it's owner, in this case bandit20 instead of the user executing it, or bandit19, ./bandit20-do cat /etc/bandit_pass/bandit20 is used to get the elevated privilege to bandit20 and passing the cat command allows the user to read a file in their base that the bandit19 account does not have permission to access which had the password.

Level 21:

```
bandit20@bandit: ~
bandit20@bandit:~$ ls -al
total 36
drwxr-xr-x   2 root     root      4096 Apr  3 15:17 .
drwxr-xr-x 150 root     root      4096 Apr  3 15:20 ..
-rw-r--r--   1 root     root       220 Mar 31  2024 .bash_logout
-rw-r--r--   1 root     root      3851 Apr  3 15:10 .bashrc
-rw-r--r--   1 root     root       807 Mar 31  2024 .profile
-rwsr-x---   1 bandit21 bandit20 15612 Apr  3 15:17 suconnect
bandit20@bandit:~$ ./suconnect
Usage: ./suconnect <portnumber>
This program will connect to the given port on localhost using TCP. If
 it receives the correct password from the other side, the next passwo
rd is transmitted back.
bandit20@bandit:~$ ./suconnect 1234
Read: 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO
Password matches, sending next password
bandit20@bandit:~$
```

```
bandit20@bandit: ~
bandit20@bandit:~$ echo -n "0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO" | nc -l -p 123
4 &
[1] 50
bandit20@bandit:~$ EeoULMCra2q0dSkYj561DX7s1CpBuOBt
bandit20@bandit:~$
```

EeoULMCra2q0dSkYj561DX7s1CpBuOBt

```
bandit21@bandit: ~

PS C:\Users\Owner\CLionProjects\CIS-4360_Computer-and-Information-Security\Assignment 3> ssh bandit21@bandit.labs.overthewire.org -p 2220
                      _                     _ _ _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-1
bandit21@bandit.labs.overthewire.org's password:

       ____..             ____          .___.
     ,'   /  \         ,/   .`|         /. ./|
    /   .     :      ,'    .'  :     .__'.  ' :
   .   /   ;.  \   ;    ;     /    /__./ \ : |
  .   .   /  \ . .'___,/    ,' .__'.  '    \' .
  ;   |  ; \ ; | |     :    | /___/ \ |    ' '
  |   :  | ; | ' ;    |.';  ; ;    \  \;      :
  .   |  ' ' ' : '____'  |  | \   ;  `      |
  '   ;  \; /  |      '   :  ;   .   \    .\  ;
   \   \  ',  /       |   |  '    \   \   ' \ |
    ;   :    /        '   :  |     :   '  |--"
     \   \ .'         ;   |.'       \   \ ;
  www. `---` ver      '---' he       '---" ire.org


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discord or IRC.

--[ Playing the games ]--

  This machine might hold several wargames.
  If you are playing "somegame", then:

    * USERNAMES are somegame0, somegame1, ...
    * Most LEVELS are stored in /somegame/.
```

The command ls -al is used to view the files in the directory where ./suconnect is found and run to see what inputs are required for it, which revealed it needed ports, in a separate terminal echo with the old password is used to pipe the password with the flag -n being used to prevent a trailing newline character from being added to the password, this password is then piped into the nc -l -p 1234, where the -l flag tells tells the netcat to listen for infoming TCP connections with -p 1234 specifying the incoming connections to be on port 1234. The amersand & pushes this process to run in the background, in another terminal the executable ./suconnect 1234 runs the setuid executable that is provided by the challenge and instructs it to connect to the custom port 1234 that was set up on the listening side where netcat is listening, the background netcat then send the password, where the suconnect bonary verified it, and returns the next level's password.